# Appendix A
# Symbols and Units

Table A.1 gives many of the physical constants used in this text. The names of general variables are provided in Table A.2, along with their units. Table A.3 lists coefficients and parameters. Some more specific parameters and constants with acronyms, are provided in Table A.4. Most variables and parameters are defined locally in the chapters, sometimes a bit differently than in these tables when there is an overlap in the use of symbols. For example, in most of Chap. 8 the flow or vascular resistance is called $R_{\text{flow}}$ (Table A.3) to avoid confusion with $R$ used for radius (Table A.2). Elsewhere where there can be no confusion it is called $R$ (and is locally defined as such).

**Table A.1** Physical constants

| Parameter (variable) | Value (in SI units) |
|---|---|
| Avogadro's number ($N_{\text{A}}$) | $6.02 \times 10^{23}$ (per mole) |
| Boltzmann constant ($k_{\text{B}}$) | $1.381 \times 10^{-23}$ J/K |
| Coulomb's Law constant ($k = 1/4\pi\epsilon_0$) | $8.99 \times 10^{9}$ N-m$^2$/C$^2$ |
| Electric permittivity ($\epsilon_0$) | $8.854 \times 10^{-12}$ F/m |
| Elementary charge ($e$) | $1.602 \times 10^{-19}$ C |
| Gas constant ($R = N_{\text{A}}k_{\text{B}}$) | 8.315 J/mole-K |
| Gravitation constant ($g$) | 9.8 m/s$^2$ = 32.2 ft/s$^2$ |
| Magnetic permeability ($\mu_0$) | $4\pi \times 10^{-7}$ N/A$^2$ |
| Planck's constant ($h$) | $6.626 \times 10^{-34}$ J-s |
| Speed of light ($c$) | $3.0 \times 10^{8}$ m/s |
| Stefan-Boltzmann constant ($\sigma$) | $5.67 \times 10^{-8}$ W/m$^2$-K$^4$ |

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**Table A.2** General variables and units

| Parameter (variable) | Common units (definition) |
|---|---|
| Acceleration ($a$, $a_{\text{decel}} = -a$) | m/s$^2$ |
| Angular momentum ($L$) | kg-m$^2$/s |
| Area ($A$, in flow $S$) | m$^2$ |
| Body height ($H$ or $H_{\text{b}}$) | m[a] |
| Body mass ($m_{\text{b}}$) | 1 kg = 1,000 g |
| Body weight ($W_{\text{b}}$) | N[b] |
| Charge ($q$) | coulombs (C) |
| Charge, number of elementary charges ($Z$) | unitless, $q = Ze$ |
| Current ($I_{\text{elect}}$, $I$) | amps (A), 1 A = 1 C/s |
| Current density ($J_{\text{elect}}$, $J$) | A/m$^2$ |
| Density (mass) ($\rho$) | 1,000 kg/m$^3$ = 1 g/cm$^3$ |
| Density (number) ($n$) | #/m$^3$, #/cm$^3$ |
| Diameter ($d$, $D$) | 1 m = 100 cm = 3.28 ft |
| Dipole moment ($P$) | 1 Debye (D) = 3.33610$^{-30}$ C-m |
| Distance ($L$), height ($h$, $y$) | m[a] |
| Electric field ($E$) | V/m |
| Energy ($E$) | J[c] |
| Flux (particle) ($J$) | #/m$^2$-s |
| Focal length ($f$) | m |
| Force ($F$, $M$ (for muscle)) | N[b] |
| Frequency (in space) ($k$) | 1/m |
| Frequency (in time) ($f$, $F$, $\nu$) | Hz, cycles per second (cps) |
| Frequency (in time) (radial) ($\omega$) | radians per second, $\omega = 2\pi f$ |
| Heat flow, amount ($Q$) | J[c] |
| Heat flow, rate ($\mathrm{d}Q/\mathrm{d}t$) | kcal/h, W[d] |
| Intensity (acoustic, optical) ($I$) | W/m$^2$ |
| Intensity (acoustic reference) ($I_{\text{ref}}$) | 10$^{-12}$ W/m$^2$ |
| Kinetic energy (KE) | J[c] |
| Loudness ($L_{\text{p}}$, $L_{\text{s}}$) | phons, (10.64); sones, (10.65) |
| Mass ($m$) | 1 kg = 1,000 g |
| Magnetic field ($B$) | 1 T (T) = 10$^4$ gauss (G) |
| Magnification ($M$) | unitless |

(continued)

**Table A.2** (continued)

| Parameter (variable) | Common units (definition) |
|---|---|
| Mobility ($\mu$) | $m^2$/V-s |
| Normal force ($N$) | N[b] |
| Osmotic pressure ($\Pi$) | Pa[e] |
| Potential energy (PE) | J[c] |
| Power ($P_{power}$ or $P$, mechanical, metabolic) | W[d] |
| Pressure ($P$) | Pa[e] |
| Radiation flux ($R$) | $W/m^2$ |
| Radius (radius of curvature) ($r$, $R$) | m[a] |
| Reaction force ($R$) | N[b] |
| Reflection coefficient ($R_{refl}$, $R$) | unitless |
| Refractive power ($P$) | 1/m = 1 D (Diopter) |
| Speed, angular, rotational ($\Omega$) | rad/s |
| Speed, velocity ($v$), flow ($u$, $v$) | m/s[f] |
| Strain ($\epsilon$) | unitless, mm/mm |
| Stress ($\sigma$) | Pa[e] |
| Temperature ($T$) | $T$(K) = $T$(°C) + 273° |
| Tension ($T$) | N[b] (for force, as in Chap. 5) |
| Tension ($T$) (surface tension) | N/m (force/length (7.4)) |
| Torque ($\tau$) or moment ($M$) | N-m |
| Transmission coefficient ($T_{trans}$, $T$) | unitless |
| Volume ($V$ or $V_{flow}$) | 1 L = 1,000 mL = 1,000 $cm^3$ |
| Volume flow rate ($Q$) | 1 L/s = 1,000 mL/s = 1,000 $cm^3$/s |
| Voltage, potential difference ($V_{elect}$, $V$) | volts (V) |
| Wavelength ($\lambda$) | m, 1 nm = $10^{-9}$ m |
| Work ($W$) | J[c] |
| Vergence ($U$ and $V$) | 1/m = 1 D (Diopter) |

Also see Fig. D.1

[a] 1 m = 100 cm = 3.28 ft, 1 mile = 5,280 ft

[b] 1 N = $10^5$ dynes = 0.225 lb, (Table 2.5)

[c] 1 J = 0.239 cal = 0.000948 BTU, 1 kcal = 4,184 J

[d] 1 W = 0.86 kcal/h = 1/746 hp = 0.00134 hp (horsepower) (Table 6.1)

[e] 1 Pa = 1 $N/m^2$, 1 MPa = 1 $N/mm^2$ = 7,600 mmHg = 10,300 $cmH_2O$ = 10 bar = 9.87 atm. Table 2.6

[f] 1 m/s = 3.6 km/h = 3.28 f/s (fps, feet per second) = 2.24 mph (miles per hour, 1 mile = 5,280 ft)

**Table A.3** General coefficients and parameters, and units

| Parameter (variable) | Common units (definition) |
|---|---|
| Absorption coefficient (sound, light) ($\gamma$) | 1/m |
| Activity factor ($f$) | unitless |
| Admittance ($Y = 1/Z = G + \mathrm{i}B$) | 1 mho = 1/ohm |
| Area moment of inertia ($I_\mathrm{A}$) | $m^4$, (4.38) |
| Capacitance ($C_\mathrm{elect}$ or $C$); per unit length | farads (F) = C/V; F/m |
| Capacitance per area ($c$) | $F/m^2$ |
| Compliance ($C_\mathrm{flow}$ or $C$); per unit length ($c_\mathrm{flow}$) | $cm^3$/bar, L/mmHg; $cm^2$/bar |
| Conductance (electrical, $G$) | siemens, 1 S = 1/ohm |
| Conductance per unit area ($g$) | 1/ohm-$m^2$ |
| Conductivity ($\sigma$) | 1/(ohm-m), $\sigma = 1/\rho$ |
| Dashpot constant ($c$) | N-s/m |
| Dielectric constant ($\kappa$) | unitless |
| Diffusion coefficient ($D_\mathrm{diff}$) | $m^2$/s, $cm^2$/s |
| Distensibility ($D_\mathrm{flow}$) | 1/Pa, (7.47) |
| Drag coefficient ($C_\mathrm{D}$) | unitless |
| Efficiency ($\epsilon$) | $0 \leq \epsilon \leq 1$ |
| Emissivity ($\epsilon$) | $0 \leq \epsilon \leq 1$, 1 for a black body |
| Friction coefficient (static, kinetic) ($\mu_\mathrm{s}$, $\mu_\mathrm{k}$) | unitless |
| Heat capacity ($C$) | kcal/°C, 1 MJ/K = 239 kcal/K |
| Heat transfer coefficient ($h = K/d = 1/I$) | W/$m^2$-°C, kcal/$m^2$-h-°C |
| Impedance ($Z = R + \mathrm{i}X$) | ohm |
| Index of refraction ($n$) | unitless |
| Insulation ($I = 1/h = d/K$) | $m^2$-°C/W, $m^2$-h-°C/kcal |
| Lift coefficient ($D_\mathrm{lift}$) | unitless |
| Moment of inertia ($I$) | kg-$m^2$, (3.23), (3.24) |
| Poisson's ratio ($\upsilon$) | unitless, (4.7) |
| Radius of gyration ($\rho$) | m |
| Reactance ($X$) | ohm |
| Resistance (flow, vascular, $R_\mathrm{flow}$ or $R$) | mmHg-s/$cm^3$ [a] |
| Resistance (electrical, $R_\mathrm{elect}$ or $R$) | ohm ($\Omega$) |
| Resistance (electrical; per unit length $r$) | ohm/m |
| Resistivity ($\rho$) | ohm-m |
| Scattering coefficient ($\alpha_\mathrm{light\ scattering}$) | 1/m |
| Skin friction coefficient ($C_\mathrm{sf}$) | unitless |
| Specific heat ($c$) | kcal/kg-°C[b] |
| Specific heat ratio ($\gamma$) | unitless, $= c_\mathrm{p}/c_\mathrm{v}$ |
| Speed of sound ($v_\mathrm{s}$) | m/s |

(continued)

**Table A.3** (continued)

| Parameter (variable) | Common units (definition) |
|---|---|
| Spring constant ($k$) | N/m |
| Stroke volume ($V_{\text{stroke}}$) | $1\,\text{L} = 1{,}000\,\text{mL} = 1{,}000\,\text{cm}^3$ |
| Surface tension ($\gamma$) | $1\,\text{N/m} = 1{,}000\,\text{dynes/cm}$ |
| Susceptance ($B$) | 1 mho = 1/ohm |
| Thermal conductivity ($K$) | W/m-K |
| Total volume flow rate ($Q_t$) | $1\,\text{L/s} = 1{,}000\,\text{mL/s} = 1{,}000\,\text{cm}^3/\text{s}$ |
| Viscosity coefficient (dynamic, absolute) ($\eta$) | Pa-s[c] |
| Viscosity coefficient (kinematic) ($\upsilon = \eta/\rho$) | Pa-s/(kg/m$^3$) |
| Young's modulus ($Y$), elastic modulus ($E$) | Pa, $1\,\text{MPa} = 1\,\text{N/mm}^2$ |

Also see Fig. D.1
[a] $1\,\text{mmHg-s/cm}^3 = 1$ mmHg-s/mL = 1 PRU
[b] per mass or volume, kcal/kg-°C, 1 MJ/m$^3$-K = 239 kcal/m$^3$-K
[c] 1 Pa-s (Poiseuille, PI) = 1 (N/m$^2$)s = 1 kg/m-s = 10 poise (P) = 1,000 cP

**Table A.4** Acronyms, including those of parameters, and units

| Parameter (variable) | Common units |
|---|---|
| Adenosine triphosphate, diphosphate (ATP, ADP) | Fig. 6.3 |
| Basal metabolic rate (BMR) | kcal/h[a] |
| Body mass index (BMI $= m_b/H^2$), Quételet's index ($Q$) | kg/m$^2$ |
| Center of mass (CM) | |
| Chromatic aberration (CA) | |
| Coefficient of restitution (COR) | unitless; (3.109) |
| Electrocardiogram (EKG, ECG) | |
| Electron transfer system (ETS) | |
| Focal length, back, effective, front (BFL, EFL, FFL) | m |
| Focal point, plane (first: F, **F**; second: F′, **F**′) | |
| Forced expiratory volume (FEV) | L |
| Functional residual capacity (FRC) | L |
| Gadd Severity Index (GSI) | s; (3.115) |
| Head Injury Criterion (HIC) | s; (3.117) |
| Inspiratory, expiratory reserve volume (IRV, ERV) | L |
| Intraocular pressure (IOP) | Pa, mmHg[b] |
| Left atrium, ventricle (LA, LV) | |
| Metabolic equivalent (MET) | unitless |
| Metabolic rate (MR) | kcal/h[a] |

(continued)

**Table A.4** (continued)

| Parameter (variable) | Common units |
|---|---|
| Near point, far point (NP, FP) | m |
| Nodal point (first: N; second: N′) | |
| Peripheral resistance unit (PRU) | mmHg-s/cm$^3$ |
| Phosphocreatine (PCr) | |
| Physiological cross-sectional area (PCA) | 1 cm$^2$ = 0.155 in$^2$ |
| Principal point, plane (first: P, **P**; second: P′, **P**′) | |
| Residual volume (RV) | L |
| Respiration exchange ratio (RER) | Unitless, Table 6.2 |
| Reynolds number (*Re*) | Unitless; (7.11) |
| Right atrium, ventricle (RA, RV) | |
| Specific stature ($S = H/m_b^{1/3}$), Ponderal index | m/kg$^{1/3}$ |
| Spherical aberration (SA) | |
| Strouhal frequency, number (*St*) | Unitless; (7.60) |
| Tidal volume (TV) | L |
| Total lung capacity (TLC) | L |
| Total peripheral vascular resistance (TPVR) | mmHg-s/cm$^3$ |
| Transient ischemic attack (TIA) | |
| Ultimate bending stress (UBS) | Pa[b] |
| Ultimate compression stress (UCS) | Pa[b] |
| Ultimate strain, ultimate percent elongation (UPE) | Unitless |
| Ultimate tensile stress (UTS) | Pa[b] |
| Visual acuity (VA) | Unitless |
| Vital capacity, forced vital capacity (VC, FVC) | L |

[a] 1 kcal/h = 1.162 W, 1 W = 0.86 kcal/h = 1/746 hp = 0.00134 hp (horsepower) (Table 6.1)
[b] 1 MPa = 1 N/mm$^2$, 1 Pa = 1 N/m$^2$, 1 MPa = 1 N/mm$^2$ = 7,600 mmHg = 10,300 cm$H_2O$ = 10 bar = 145 psi = 9.87 atm. Table 2.6